

Taimur Shahzad Gill

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RESEARCH INTERESTS

I am a computational biologist interested in how transcriptomic data can be resolved into the cell types and pathways that shape phenotype. My MSc dissertation at the Earlham Institute benchmarks cell type deconvolution and single-cell workflows for the *in silico* prediction of pathogenic cell types and pathways in Crohn's disease, integrating bulk and single-cell RNA-seq with clinical metadata across large public cohorts. My wider interest is in methods development and benchmarking for deconvolution and multi-omics integration, and in machine learning that remains interpretable when it is used to make biological claims. I care about reproducible, openly shared pipelines that let these methods transfer between systems and species.

EDUCATION

MSc Data Science for Biology

Sep 2025 – Present

University of East Anglia (Norwich, United Kingdom)

- Dissertation: *In silico* prediction of pathogenic cell types and pathways in Crohn's disease, supervised by Prof Irene Papatheodorou and Dr Gregory Wickham at the Earlham Institute.
- Modules: Data Science and Bioinformatics; Statistics for Biologists; Data Mining.

BEng Electrical Engineering

Sep 2019 – Jun 2023

National University of Sciences and Technology (NUST) (Islamabad, Pakistan)

- Relevant coursework: Machine Learning (A), Digital Image Processing (A), Calculus and Analytical Geometry (A), Artificial Intelligence and Decision Support Systems (B+), Digital Signal Processing (B+), Numerical Methods (B+).

RESEARCH EXPERIENCE

Visiting Student Researcher; Papatheodorou Group

Nov 2025 – Present

Earlham Institute, Norwich Research Park (Norwich, United Kingdom)

- Integrate bulk and single-cell RNA-seq with clinical metadata across large public Crohn's disease cohorts to identify pathogenic cell subtypes and disease-progression trajectories, working with GEO datasets including GSE57945, GSE93624, and GSE192786.
- Benchmark cell type deconvolution of bulk RNA-seq using CIBERSORTx, MuSiC, and Bisque within the CATD pipeline, quantifying how reference choice and method assumptions drive differences in estimated cell type proportions.
- Process single-cell data through the EISCA pipeline for quality control, normalisation, clustering, and cell type annotation, maintaining reproducible Python and R workflows under version control.
- Develop machine learning models combining deconvolved cell type proportions, cell-type-specific expression signatures, and clinical parameters to predict disease biomarkers and progression.

Honorary Research Assistant; Nixon Research Group

May 2025 – Present

Department of Mathematical Sciences, University of Liverpool (Liverpool, United Kingdom)

- Evaluated the TimeGPT foundation model for influenza surveillance forecasting, achieving an RMSE of 873.27 at a one-week horizon and outperforming traditional baselines (Python, scikit-learn, pandas).
- Led a seasonal pattern analysis of severe fever with thrombocytopenia syndrome in South Korea, using complex empirical mode decomposition, harmonic regression ($R^2 = 0.848$), and bicoherence analysis to resolve multi-scale transmission cycles (Python, R).

Data Analyst Intern (REF2029)

2025 – Present

Research and Innovation Services, University of East Anglia (Norwich, United Kingdom)

- Model how proxy indicators — research income per FTE, PhD studentships, and publication venue — relate to output-level GPA using REF 2021 and OpenAlex bibliometric data, validating measures with Unit of Assessment coordinators across disciplines and preparing the analysis for publication.

AddiTechSim, University of Stuttgart (Stuttgart, Germany)

- Delivered an independent research project applying deep reinforcement learning (Python, Stable-Baselines3, OpenAI Gymnasium) to a parameter optimisation problem, converging 98% faster than the existing Q-learning approach.

PUBLICATIONS

Peer-reviewed journal article

- Gill, T. S., Zaidi, S. S. H., & Shirazi, M. A. (2024). Attention-based deep convolutional neural network for classification of generalized and focal epileptic seizures. *Epilepsy & Behavior*, 155, 109732.

Peer-reviewed conference proceedings

- Gill, T. S. & Zahid, S. I. (2024). Time series forecasting of the KSE-100 index using a hybrid ESN-LSTM model. *2024 International Conference on Robotics and Automation in Industry (ICRAI)*.
- Gill, T. S., Shirazi, M. A., & Zaidi, S. S. H. (2023). Early detection of mesothelioma using machine learning algorithms. *Engineering Proceedings*, 46(1), 6. 7th International Electrical Engineering Conference.
- Gill, T. S., Shehwar, D. E., Memon, H., Khanam, S., Ahmed, A., Shaukat, U., Mateen, A., & Zaidi, S. S. H. (2021). IoT-based smart power quality monitoring and electricity theft detection system. *2021 16th International Conference on Emerging Technologies (ICET)*, Islamabad, Pakistan.

TECHNICAL SKILLS

Genomics and bioinformatics: bulk and single-cell RNA-seq analysis; cell type deconvolution (CIBERSORTx, MuSiC, Bisque) via the CATD pipeline; single-cell QC, normalisation, clustering, and annotation via the EISCA pipeline; multi-omics integration; retrieval and reuse of public data from GEO

Programming: Python (pandas, NumPy, scikit-learn, Scanpy, Matplotlib), R (Seurat, Bioconductor, tidyverse, ggplot2), Bash, SQL, C/C++

Machine learning and statistics: deep learning, attention mechanisms and transformers, reinforcement learning, time-series and forecasting models, harmonic regression, explainable AI (SHAP)

Reproducibility and computing: Linux, Git and GitHub, high-performance computing environments, $\text{\LaTeX}$ (Overleaf / R Markdown), open data and FAIR principles

Languages: English (fluent, IELTS 7.5/9.0), Urdu (native), Hindi (basic), Spanish (beginner)

OUTREACH AND SERVICE

- Public engagement volunteer, Microbe Zoo, Royal Society Summer Science Exhibition — communicating microbial and genomic science to general audiences.
- Co-Founder, NUST Robotics and Artificial Intelligence Society — established and grew a student research society from the ground up.
- Treasurer and Director of Publications, Education and Community Service Society (2019–2021).

REFEREES

Professor Irene Papatheodorou

Head of Data Science, Earlham Institute, Norwich Research Park – MSc dissertation supervisor;
Irene.Papatheodorou@earlham.ac.uk

Dr Emily Nixon

Lecturer in Mathematics for Healthcare, Department of Mathematical Sciences, University of Liverpool – research supervisor; Emily.Nixon@liverpool.ac.uk